PROPOSAL · STARTING STAGE

Qlik-to-SQL Reconstruction

The first stage of the Agentic Dataset Context Engine: reconstructing how Qlik reports are produced, as validated SQL on Dremio.

Document	Starting-stage proposal
Part of	Agentic Dataset Context Engine (see the overall proposal)
Prepared for	Data & Engineering Leadership
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1. Summary

Many reports in the organization are produced by Qlik applications whose logic lives in load scripts, intermediate QVD files and chart expressions. That logic is hard to audit, hard to migrate, and rarely documented. The data behind it now lives in Dremio.

We propose to start the Agentic Dataset Context Engine with the capability that proves its central claim: **reconstructing how reported numbers are produced**. The starting stage reads Qlik scripts, translates their logic into **Dremio SQL**, runs that SQL in Dremio, and checks the results against independent evidence (reload logs, QVD files and exported reports). Every result carries an explicit validation level, so nothing is presented as correct without proof.

Everything the starting stage builds is part of the final product: the knowledge graph, the review workflow, the Dremio connection and the model gateway are built once, in their final shape, and extended in later stages.

> **Ask: approve the 12-week starting stage on 2–3 real Qlik applications, with read-only access to Dremio and the Qlik artifacts listed in section 10.**

2. Why start with Qlik

Reason	What it means
It proves the headline	Reconstructing report logic is the product's main promise. Starting here proves it in the first weeks rather than at the end.
The inputs exist today	Qlik scripts are available now. Access to Qlik servers can come later and is not required to start.
Translation is reliable	Translating real logic is far more deterministic than inferring SQL from report outputs, so this is the lowest-risk route to a working end-to-end result.
Results can be proven	Reload logs, QVD files and exported reports give independent ground truth without needing the Qlik engine (section 5).
It seeds the knowledge base	Lineage from scripts and QVD chains, joins from Qlik data models and definitions from master items become trusted knowledge for every later stage.
It has value on its own	A validated migration pack per application supports audits, decommissioning and migration, even before the rest of the platform exists.

3. What the starting stage delivers

At the end of 12 weeks:

1. Real Qlik applications (QlikView and Qlik Sense) are read from their scripts, including chains of applications connected through QVD files.
2. Every Qlik table is reproduced as Dremio SQL, executed in Dremio, and validated as far as the available evidence allows.
3. Where measure definitions are available, report figures are reproduced and compared with exported reports.
4. All findings are stored as knowledge with evidence and history, and shown as a lineage map from Dremio datasets through Qlik tables and QVDs to report visuals.

5 Ambiguities (unresolved variables, uncertain source mappings, statements that could not be translated) are resolved by reviewers in a workbench, and their decisions are reused.

6 A migration pack per application can be exported: SQL, lineage, a validation report and a list of open items.

7 The same build runs locally and in a staging environment, ready to move into production and into client environments later.

4. How it works

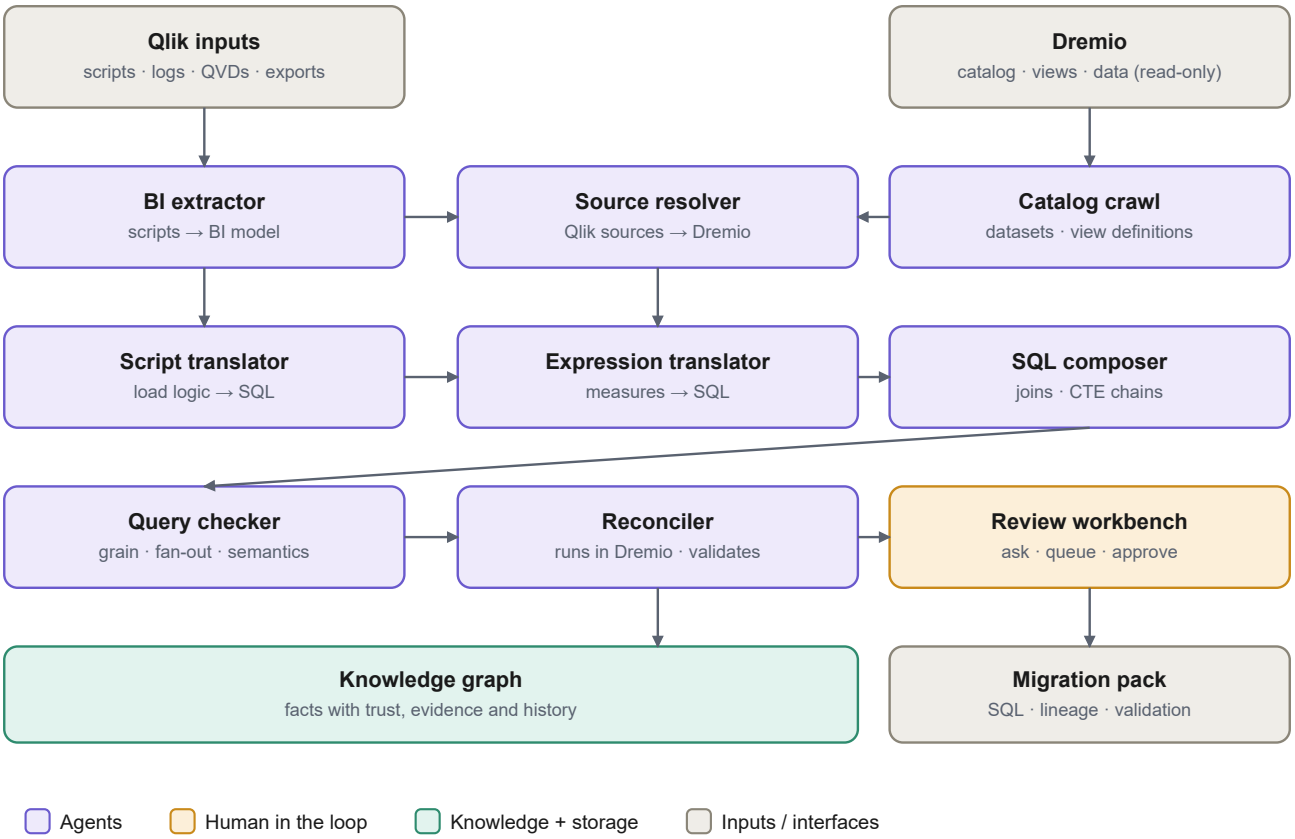


Figure 1. The starting-stage pipeline. Qlik artifacts and Dremio metadata are combined; every result is recorded as knowledge and reviewed before it is trusted.

Step	What happens
1. Intake	Scripts, QlikView project folders, reload logs, QVD files and exported reports are uploaded and classified per application.
2. Extract	Scripts are split into statements and parsed into a tool-neutral model of the application: tables, transformations, variables, data model and measures. Nothing is silently dropped; anything not understood is flagged.
3. Resolve sources	Each source a script reads is mapped to a Dremio dataset or to the Qlik application that produced the QVD file. Uncertain mappings go to review.
4. Translate	Embedded source SQL is converted to the Dremio dialect, and Qlik transformations (joins, lookups, concatenations, set analysis) become SQL. AI assistance is used only for statements the rules do not cover, and its output is always re-checked.
5. Check	Generated SQL is checked for known Qlik-to-SQL pitfalls, most importantly double counting caused by differences in how Qlik and SQL aggregate across joined tables.

Step	What happens
6. Execute and validate	SQL runs in Dremio under strict load limits, and results are compared with reload logs, QVD files and exported reports (section 5).
7. Review and record	Reviewers confirm mappings, answer questions and approve SQL. Every decision is recorded with its evidence.
8. Export	A migration pack is produced per application (section 6).

5. How results are validated

Each reproduced table or figure is labelled with the **highest level of evidence it reaches**. Results are never described as correct without a level.

Level	What is checked	Evidence used
V0 Parsed	Every script statement is accounted for	Script text
V1 Checked	SQL is valid for Dremio and passes the semantic checks	Rules and knowledge graph
V2 Executes	SQL runs in Dremio and returns the expected columns	Dremio
V3 Row counts match	Rows per table match what Qlik loaded	Reload logs, QVD file headers
V4 Data matches	Column-level totals, distinct counts and ranges match	QVD file contents
V5 Figures match	Report figures match what users see	Exported reports or owner-provided figures
V6 Engine match	Results match the live Qlik engine across several filter selections	Qlik server (when access is available)

Point-in-time alignment. Qlik evidence reflects data at the time of the last reload, while Dremio returns current data. Where Dremio tables support time travel, queries run as of the reload time and comparisons are exact. Otherwise differences are reported with a stated tolerance and the time gap is recorded as the reason; such results are never labelled exact.

6. The deliverable: a migration pack

For each Qlik application, a versioned pack that can be regenerated identically from the same inputs:

Part	Contents
SQL	Dremio SQL for every Qlik table and, where definitions exist, for every measure. Producer applications in QVD chains are included, so the SQL reads only Dremio datasets.
Lineage	A map from Dremio datasets to Qlik tables, QVD files, applications and report visuals, in a readable summary and in the open OpenLineage format.
Validation report	The level reached by each table and figure, with differences, tolerances and their reasons.
Review record	Statements that could not be translated, reviewer decisions applied, and open items such as a missing producer application.
Manifest	Which inputs were received and which software and model versions produced the pack.

7. Data protection and environments

- **Read-only, always.** The system reads Dremio and Qlik artifacts. It never writes to Dremio, never reloads or changes Qlik applications.
- **No row-level data reaches an AI model in this stage.** Models see scripts, schemas, view definitions and aggregate differences only. This is enforced centrally in the model gateway.
- **Controlled load on Dremio.** Validation queries are bounded, cached, labelled so their load is visible, limited in concurrency and scheduled off-peak where needed.
- **Models through Google Cloud.** AI models are accessed through Google Cloud (Vertex AI) in an approved project and region. Client deployments can use the client's own approved endpoint.
- **Local first, then production.** The work starts in local environments with a local copy of sample data, moves to a staging environment on Google Cloud, and is packaged so it can later be installed in production or inside a client's own environment.
- **Full audit trail.** Every finding, decision, prompt version and model version is recorded.

8. Plan and milestones

Phase	Weeks	Build	Done when
Foundations	1–2	Local environment with Dremio and sample data; shared data formats; knowledge store; model gateway; first Qlik script parser on every sample application; readers for reload logs and QVD files	Sample applications are parsed with measured coverage, and a first Qlik table runs as SQL in Dremio
Tables	3–5	Complete script extraction and translation; source resolution including QVD chains; Qlik-to-SQL checks; validation against reload logs and QVD files; review workbench	At least 90% of script statements translated and at least 80% of tables matching on row counts (V3 or higher)
Figures and lineage	6–8	Measure translation; reading exported reports; composed SQL per visual; lineage map; figure-level validation	At least three real report figures reproduced and matching exported reports (V5)
Harden and package	9–12	More applications; complex constructs; migration pack export; performance and load tuning; staging environment	Targets in section 9 met; a migration pack accepted by a report owner; the same build running in staging

Progress is reviewed at the end of each phase against these criteria. In parallel, from week 6, the wider platform (profiling, dataset matching, data dictionary and estate analysis) starts building on the same foundations; see the overall proposal.

9. Success measures

Measure	Target by week 12
Script statements translated without being flagged	At least 90% per application
Tables with matching row counts (V3 or higher)	At least 80%
Tables with matching data where QVD files exist (V4)	At least 70%
Report figures matching exported reports (V5), for standard constructs	At least 70%
Results shown with their validation level	100%; no unlabelled "match"
Unresolved variables or sources silently assumed	None; all go to review
Migration packs regenerated identically from the same inputs	100%

Measure	Target by week 12
Cost per application translated	Measured and reported

10. What we need to start

Input or access	Why it is needed
2–3 representative Qlik applications: load scripts, QlikView project folders or app files	The material the pipeline reads, and the basis of the test set
Reload logs for those applications	Resolve variables and confirm row counts (V3)
QVD files they read and write	Lineage across applications and data-level checks (V3, V4)
Exported reports or reference figures for 10 or more visuals	Figure-level validation (V5)
Read-only Dremio access, or a sample dump for local work	Source resolution, execution and validation
Names of the Qlik data connections and what they point to	Mapping Qlik sources to Dremio
A Google Cloud project with Vertex AI access	Model access for translation assistance
Review time from a report or data owner	Confirm mappings and approve reconstructed SQL

11. Risks and mitigations

Risk	Mitigation
Unusual Qlik syntax the parser does not yet handle	The grammar grows from real scripts; anything not understood is flagged, never dropped; coverage is measured per application from week 1
Only load scripts are provided, without measure definitions	Table-level SQL and lineage are still delivered; figure-level work starts as soon as project folders or app files are available
Data has changed since the last Qlik reload	Point-in-time queries where available; otherwise stated tolerances with the time gap recorded
Qlik reads sources that are not in Dremio, or reads one of several copies	Reviewed connection mapping; the reconstruction reproduces exactly what Qlik reads and flags where a better (golden) source exists
A producer application in a QVD chain is missing	The result is marked not resolved and the missing application is named in the pack
Validation queries add load to Dremio	Bounded, cached, labelled and rate-limited queries, scheduled off-peak

12. What comes next

The starting stage is the first part of the Agentic Dataset Context Engine. From week 6, the wider platform builds on the same foundations: profiling of Dremio datasets, key and join discovery, detection of overlapping and duplicate datasets with recommended golden records, the living data dictionary, and reconstruction of reports for which no Qlik logic is available. Access to the Qlik servers, when granted, adds live validation against the Qlik engine. At a later stage, everything learned is used to propose a designed data model (a star schema, a normalized business model or a semantic layer of curated Dremio views), proven by reproducing existing reports. The overall proposal describes the complete product and roadmap.

13. Decision requested

Approve the 12-week starting stage: Qlik-to-SQL reconstruction on 2–3 real Qlik applications, with read-only Dremio access, the Qlik artifacts in section 10, a Google Cloud project for model access, and an infrastructure and model budget to be sized on Google Cloud pricing in the first two weeks. Progress is reviewed at the end of each phase against section 9.